Robe variants - screenshots with settings

UO_Model3D, branch model_fixing5, October 2026. One free robe model (Meshy, 24,366 vertices after conform) on the body 0x190, rendered with the pipeline's client-style look (UO_Look). Every page is a contact sheet exactly as it was rendered during the work; the row labels are inside the images, the settings of each row are in the caption. Walk sheets: 10 frames = 5 directions x 2 frames (cycle frames 0 and 5 of each direction). Original robe 469 rows (where present) are the original client sprites for comparison. Metrics and tables: robe_variants_2026-10.pdf.

Contents: 1. Cloth physics (free cloth, pin weight, leg push, envelope, collision margin) 2. Fabrics 3. Final settings t1 per action 4. Riding 5. Fit (loose / tight / slim cuffs) 6. Shading (contrast curve, contact shadow, inner line)

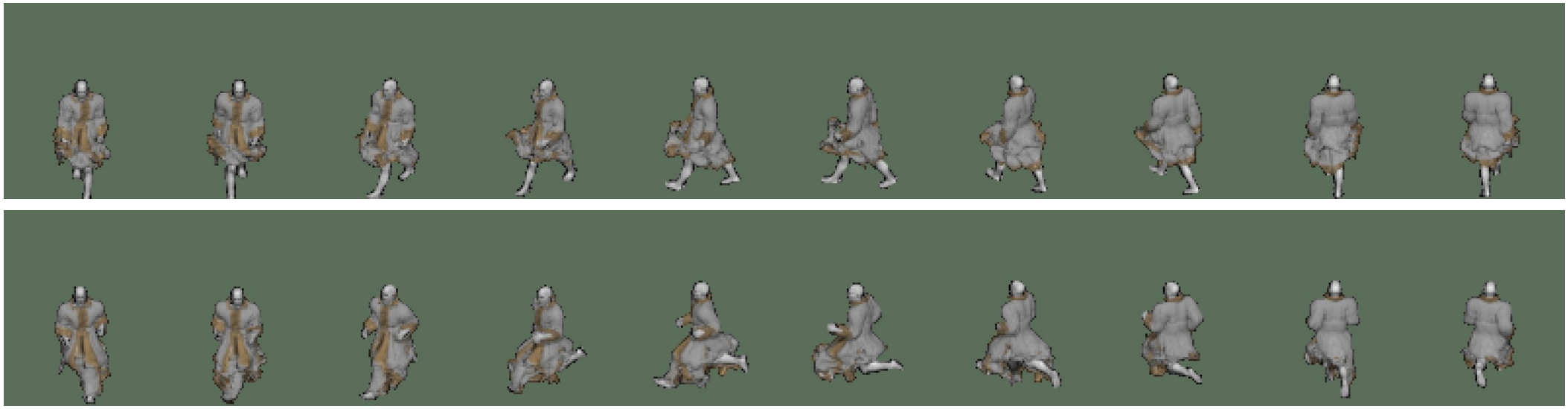
1.1 Free cloth, cotton, walk (goal 0)

Free Blender cloth, no pin (goal 0), cotton, margin 0.02 (P1/P2 of the report). The hem climbs and bunches at the waist, the legs stand out of the robe: rejected.



1.2 Cotton, free cloth, collision margin 0.006: walk and run

Cotton, goal 0, collision margin 0.006 m, mass scaled to the mesh. Margins this thin let the legs tunnel through the cloth (see leg px in the report).



1.3 Pin to the leg-pushed target: walk

Silk, walk. Rows: old (goal 0.2, push 1.0 = old hull target) | pushw 0.5 (pin target = skinned + 0.5 x leg push) | pushw 0 (hangs from the pelvis only, legs poke through the front) | pushw 0 + gravity 2 (same, more exposure). Chosen: pushw 0.5.



1.4 Pin to the leg-pushed target: spell (directed)

Same rows as 1.3 on action 16. Chosen: pushw 0.5.



1.5 Envelope (leg hull) instead of free skirt: walk

Silk, walk. Rows: old physics (hull target) | new, no envelope | E1 (envelope, margin 0.02) | E2 (envelope, margin 0.04, tent). The envelope flattens the robe into a bell-shaped skirt again: rejected.



1.6 Envelope: run

Same rows as 1.5, action 2 (run).



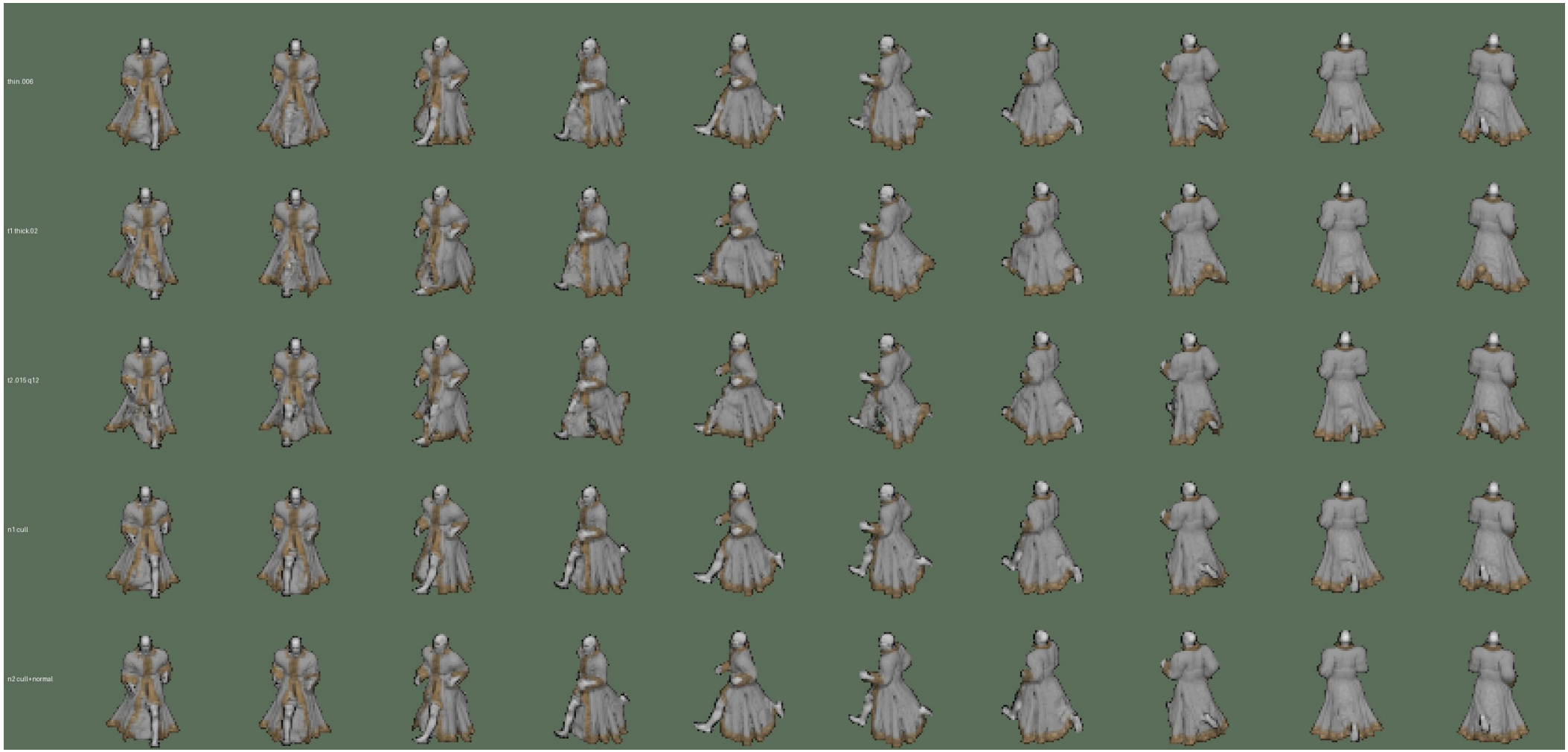
1.7 Collision settings: walk

Silk, walk. Rows: thin (thick/margin 0.006) | t1 (thick 0.02, dist 0.005, chosen) | t2 (0.015, quality 12) | n1 (cull inside-facing collision faces) | n2 (cull + use collider normals, a colleague's idea). Leg px through the cloth: thin 2396, t1 1435, t2 2118, n1 2628, n2 2642 (silk walk+run, 245 frames).



1.8 Collision settings: run

Same rows as 1.7, action 2 (run).



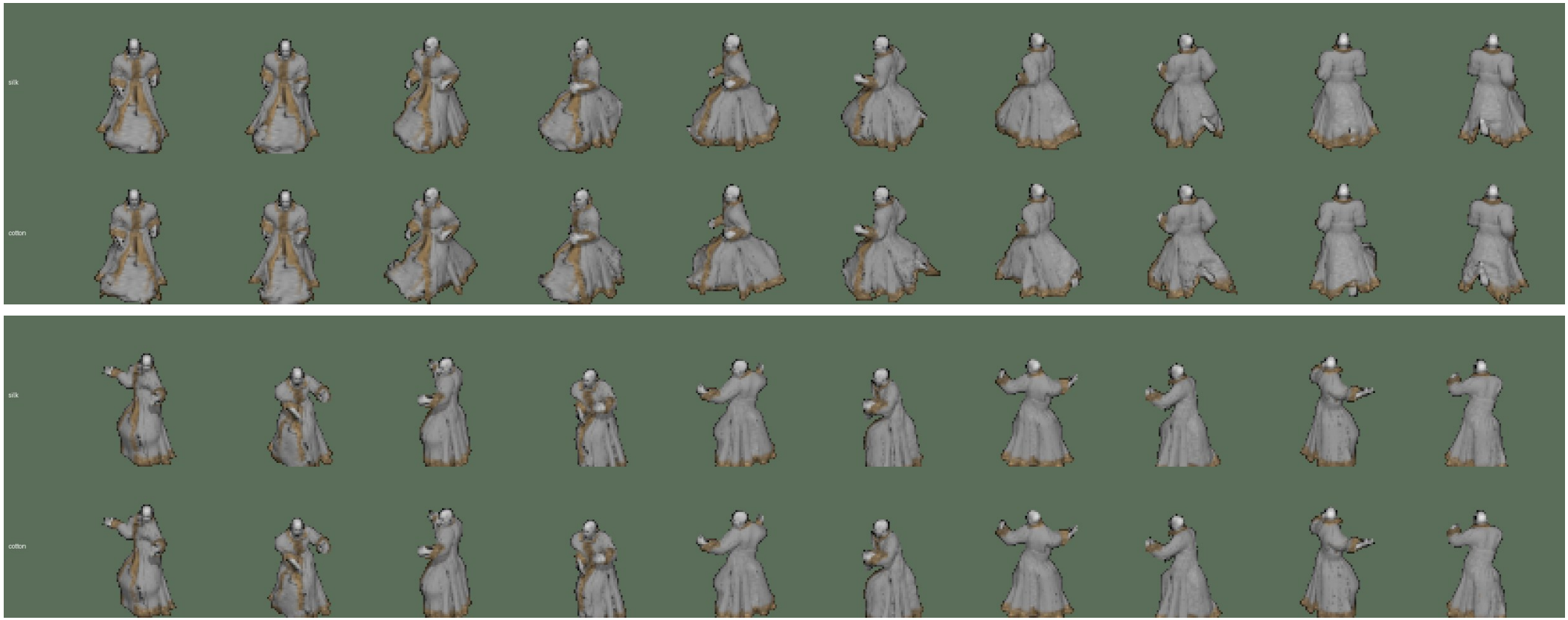
2.1 Fabrics, first presets: walk

Silk vs cotton, walk, before the in-plane stiffness fix: cotton hem lifts +0.26 m, wool +0.47 m (silk +0.04). Tension/compression/shear differed per preset.



2.2 Fabrics, first presets: run, slash

Silk vs cotton, actions 2 (run) and 9 (1h slash), same settings as 2.1.



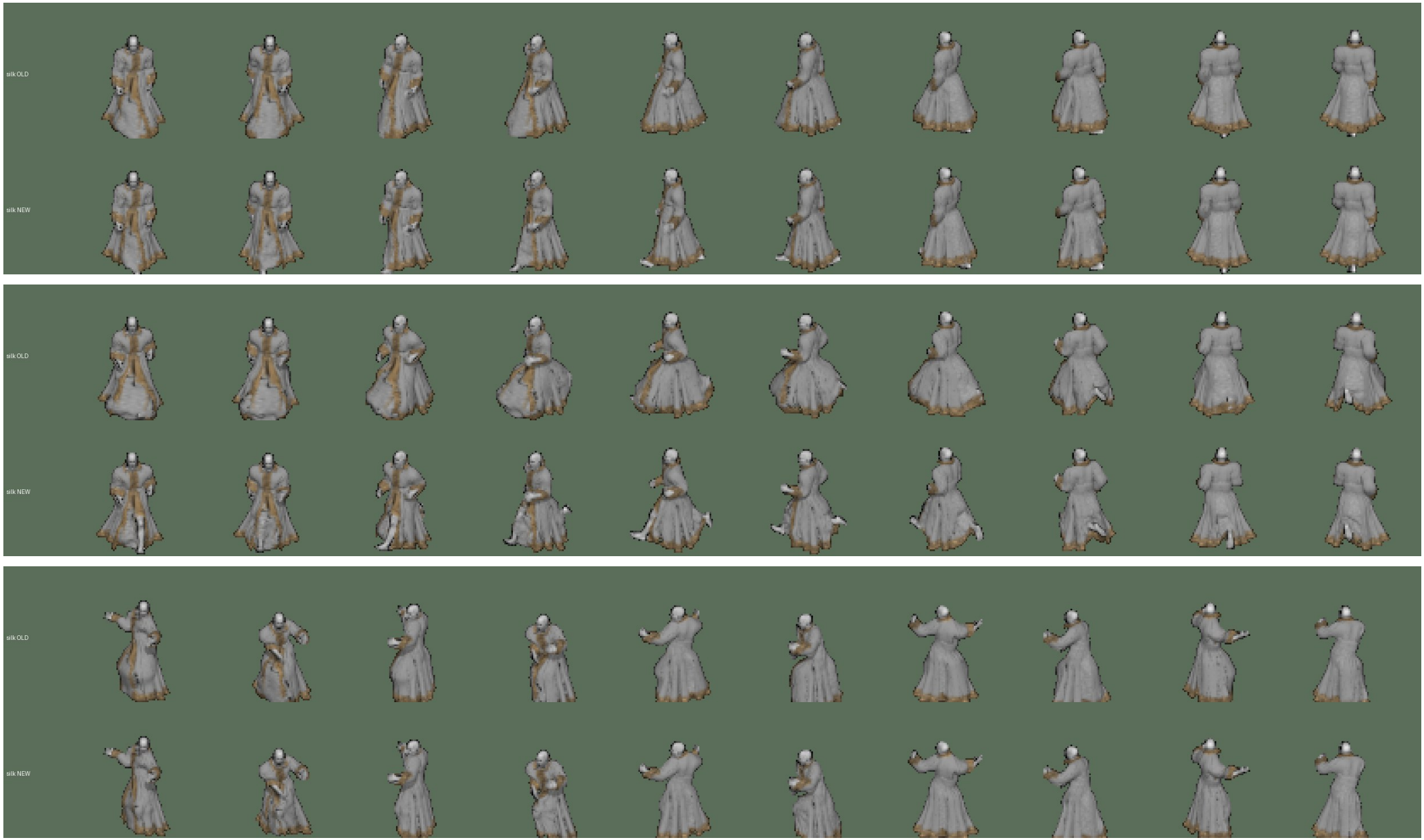
2.3 Fabrics, first presets: stand, spell (directed)

Silk vs cotton, actions 4 (stand) and 16 (spell directed), same settings as 2.1.



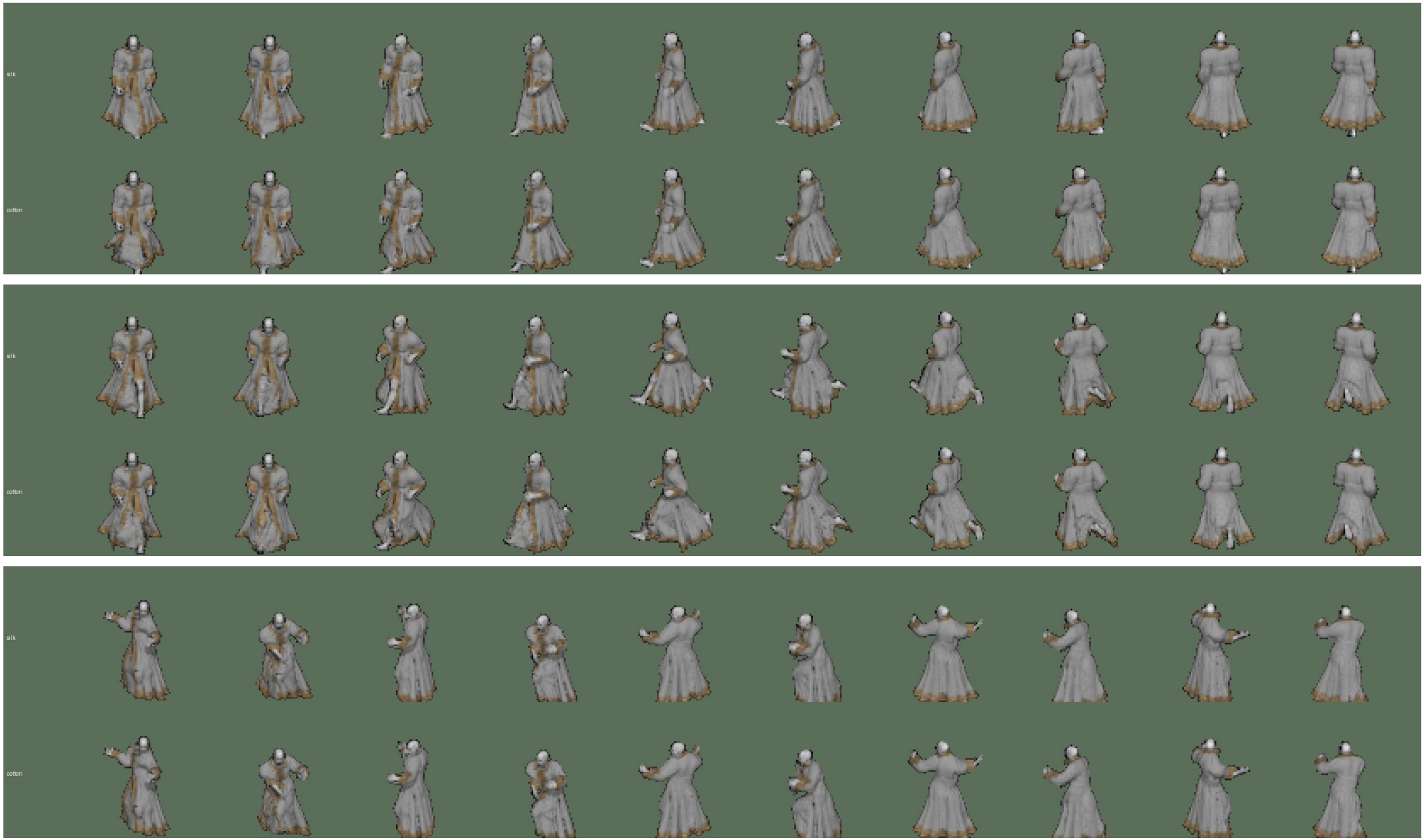
2.4 Old (hull) vs new physics, silk: walk, run, slash

Silk, old physics (hull target, goal 0.2, push 1.0) vs new (pin goal 0.2, pushw 0.5). Row labels in the images.



2.5 Silk vs cotton after the fix: walk, run, slash

Silk and cotton (mass .15 / .3, bending .05 / .5, damping 1 / 5, in-plane stiffness 5), pushw 0.5, goal 0.2, mscale 0.0333, margin 0.02. Hem lift in the walk +0.04 for both.



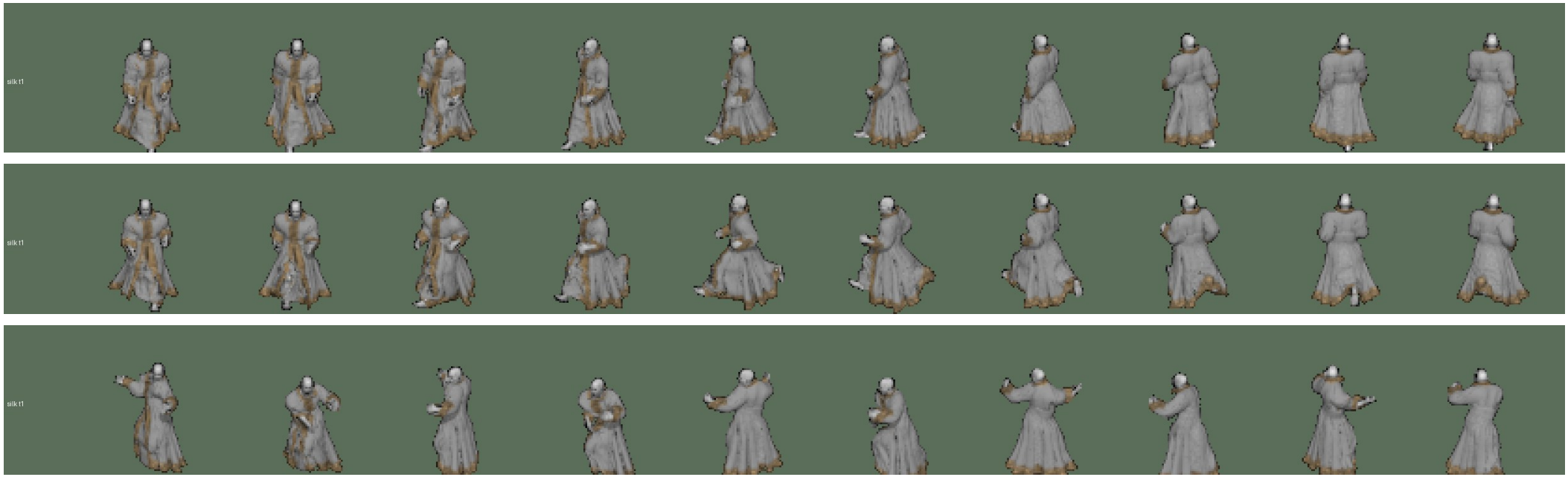
2.6 Silk vs cotton: die forward / die backward

Silk and cotton, actions 21 and 22, same settings as 2.5.



3.1 Final settings t1 (silk): walk, run, slash

t1: fabric=silk mscale=0.0333 goal=0.2 pushw=0.5 gravity=1.5 in_max=-1 max_move=0 smooth=0.04 arm_goal=0.95 preroll=20 cycles=2 thick=0.02 dist=0.005. Cotton and wool use the same keys with their fabric names.



3.2 Final settings t1 (silk): die forward / backward

Same settings as 3.1, actions 21 and 22. Leg exposure is the largest remaining problem here (die forward: 385 px through the cloth over 30 frames).



4.1 Riding, first version (mounted rest)

Mounted walk, early test: the robe hangs from the pelvis without a horse collider; it spreads like trousers.



4.2 Riding with the horse as a collider: silk and cotton

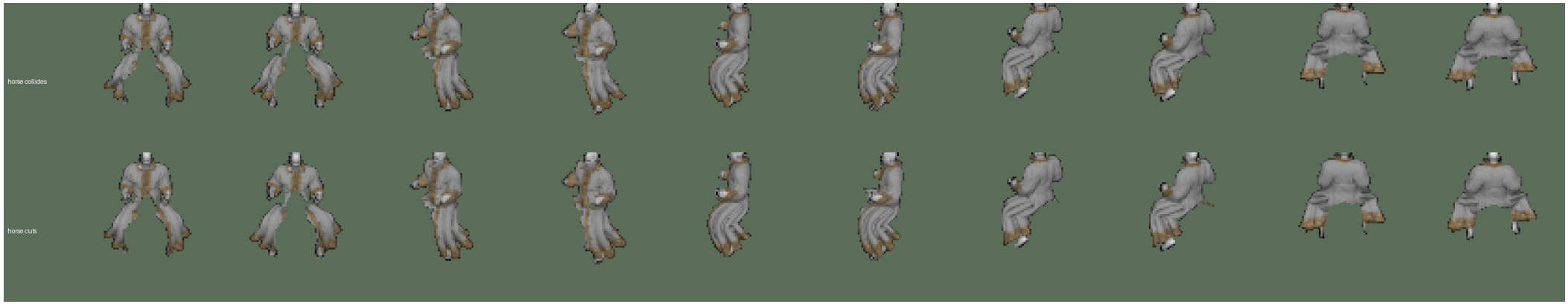
Actions 23 and 24 (mounted walk / run), t1 + mounted=1 horse=1. Rows: 'horse collides' = the horse body proxy is a cloth collider; 'horse cuts' = the horse only cuts the cloth (that variant was not made for the full VDs).





4.3 Riding, t1 final (silk): mounted walk / run

t1 with mounted=1 horse=1, silk.





5.1 Fit: loose vs tight (walk, 5 directions, 2 rows each)

Loose: conform GAP 1.5 cm, LOOSE_K 0.25 (rows 1-2). Tight: GAP 1.0 cm, LOOSE_K 0.4, FOLD_MAX 0.4, POSE_GAP 0.8 (rows 3-4), tighter from the waist up. The difference is 0.7-0.9 cm (0.3 px).



5.2 Slim cuffs vs original robe 469 (1/2)

Rows per action: original robe 469 | ours silk | slim cuffs (SLEEVE_CUFF 3.2 cm, CUFF_SPAN 0.25). Actions 17 (spell area) and 0 (walk). The sleeves are about 10% thicker than the original; slim cuffs help partly.



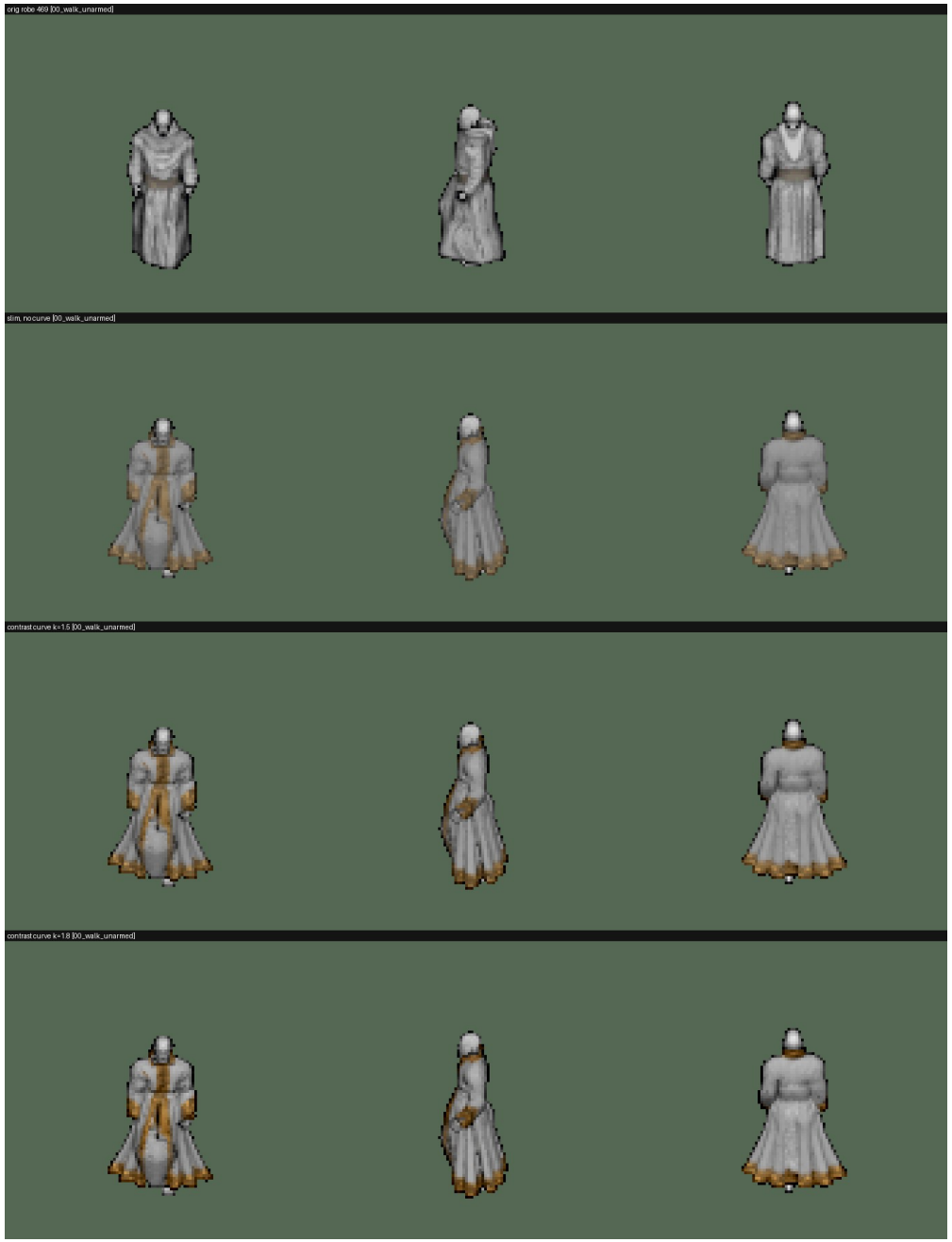
5.2 Slim cuffs vs original robe 469 (2/2)

Rows per action: original robe 469 | ours silk | slim cuffs (SLEEVE_CUFF 3.2 cm, CUFF_SPAN 0.25). Actions 17 (spell area) and 0 (walk). The sleeves are about 10% thicker than the original; slim cuffs help partly.



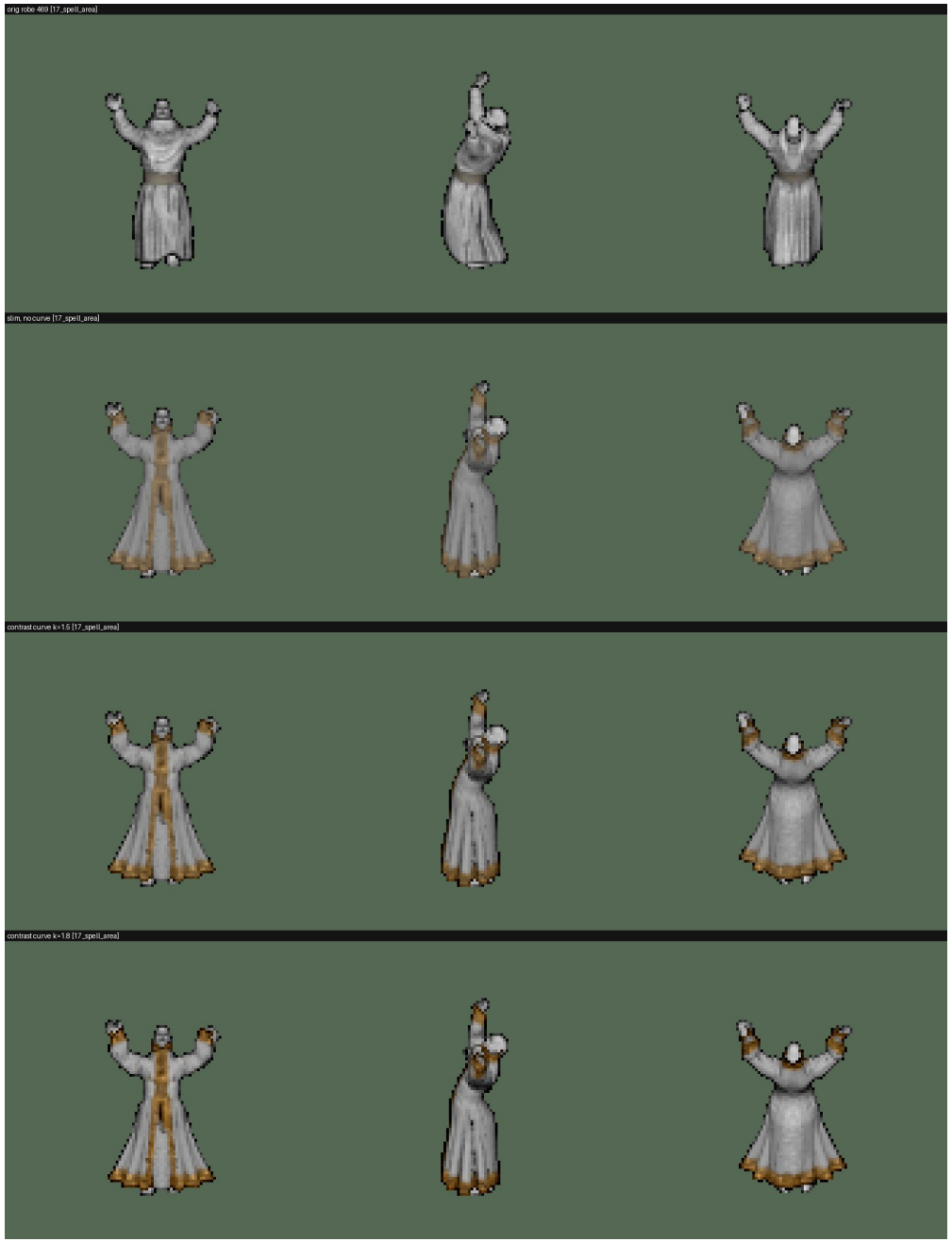
6.1 Contrast curve k 1.5 / 1.8 (1/2)

Slim silk, walk and spell area. Rows: original robe 469 | no curve | contrast curve k 1.5 | k 1.8 (offline curve on the rendered image; in the renderer: uo_contrast). Accepted by the user as an option.



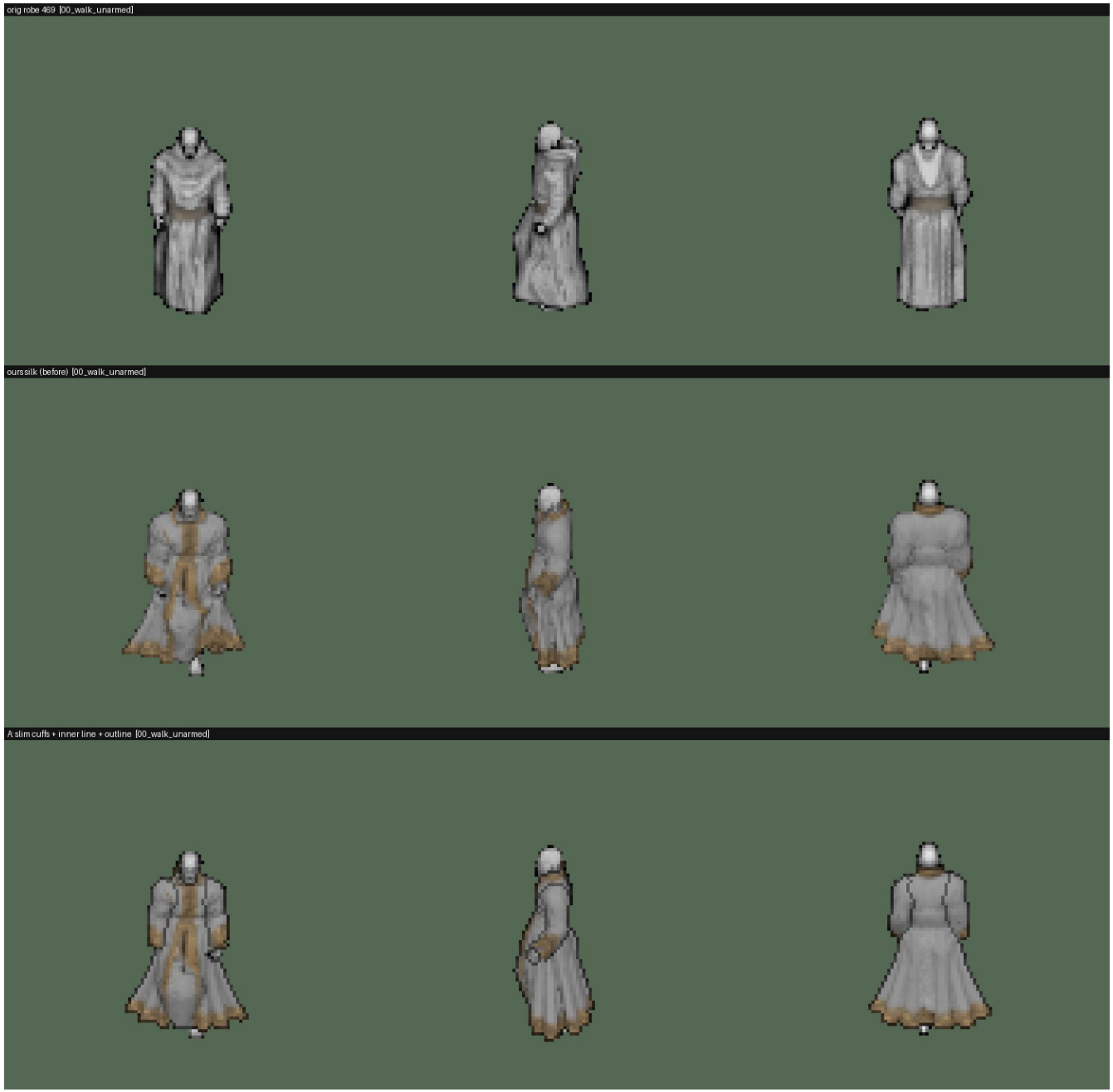
6.1 Contrast curve k 1.5 / 1.8 (2/2)

Slim silk, walk and spell area. Rows: original robe 469 | no curve | contrast curve k 1.5 | k 1.8 (offline curve on the rendered image; in the renderer: uo_contrast). Accepted by the user as an option.



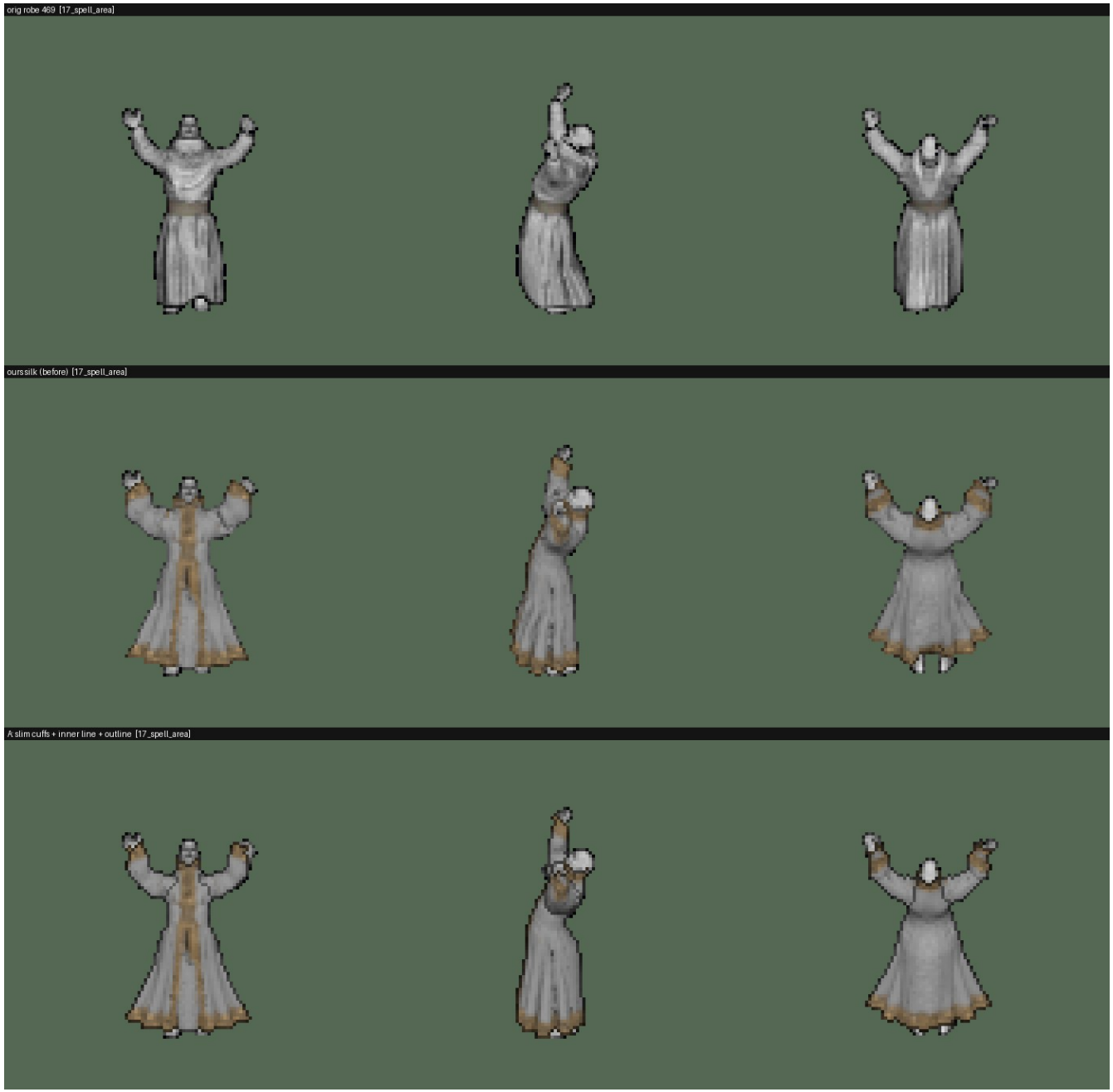
6.2 Variant A: slim cuffs + inner line + outline (1/2)

Rows: original robe 469 | ours silk (before) | A: slim cuffs + uo_inner_line + uo_outline. The line around the arm looks drawn (user: unaesthetic): rejected.



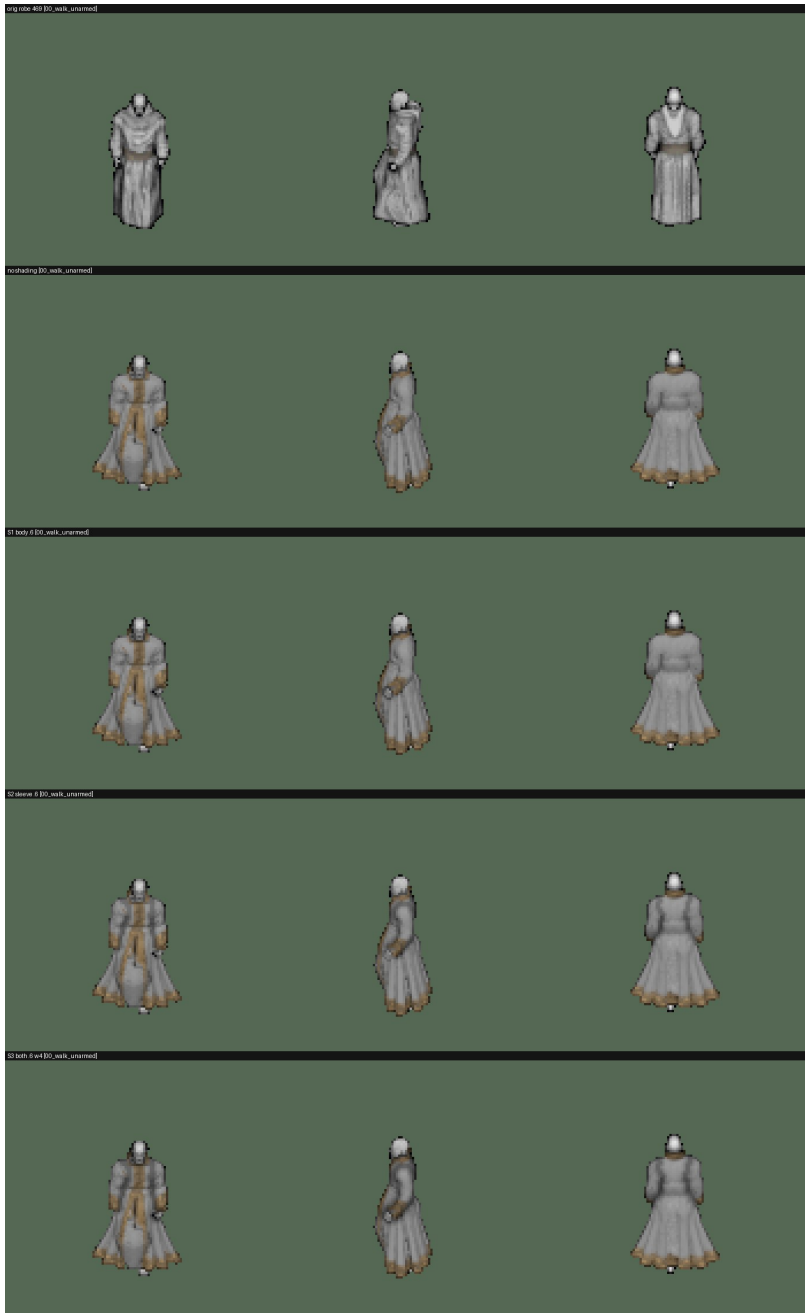
6.2 Variant A: slim cuffs + inner line + outline (2/2)

Rows: original robe 469 | ours silk (before) | A: slim cuffs + uo_inner_line + uo_outline. The line around the arm looks drawn (user: unaesthetic): rejected.



6.3 Contact shadows (uo_shade_sleeve / uo_shade_body) (1/2)

Rows: original robe 469 | no shading | S1 body shade 0.6 | S2 sleeve shade 0.6 | S3 both 0.6, width 4 px (walk and spell area).



6.3 Contact shadows (uo_shade_sleeve / uo_shade_body) (2/2)

Rows: original robe 469 | no shading | S1 body shade 0.6 | S2 sleeve shade 0.6 | S3 both 0.6, width 4 px (walk and spell area).



6.4 Contact shadows, zoom

Zoom of 6.3 on the sleeve and the arm pit.



6.5 Shading combinations T1-T4 (zoom)

Columns: original 469 | no shading | T1 shade 0.4 width 4 | T2 shade 0.5 width 4 | T3 shade 0.4 + contrast 1.4 | T4 shade 0.5 + contrast 1.5. Decision: base look of a robe/skirt = uo_contrast 1.5 + uo_shade_sleeve 0.4 + uo_shade_width 4.

